

# Translations



1 Consider the parabola  $y = x^2 - 3$ .

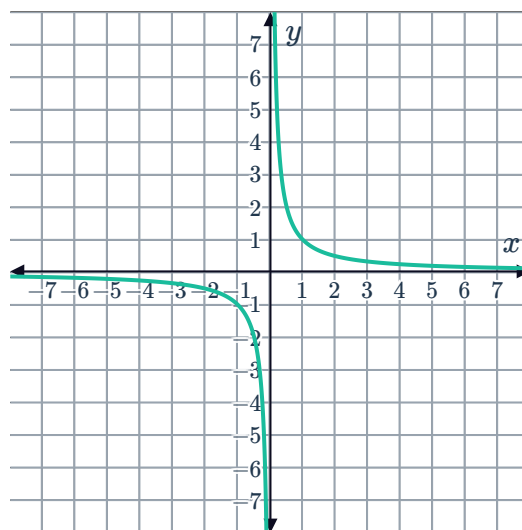
- a Complete the table of values:
- b Sketch the graph of  $y = x^2 - 3$ .
- c What is the  $y$ -intercept of the graph?

|     |    |    |   |   |   |
|-----|----|----|---|---|---|
| $x$ | -2 | -1 | 0 | 1 | 2 |
| $y$ |    |    |   |   |   |

- d When adding a constant to the equation  $y = x^2$ , describe the type of transformation that occurs on its graph.

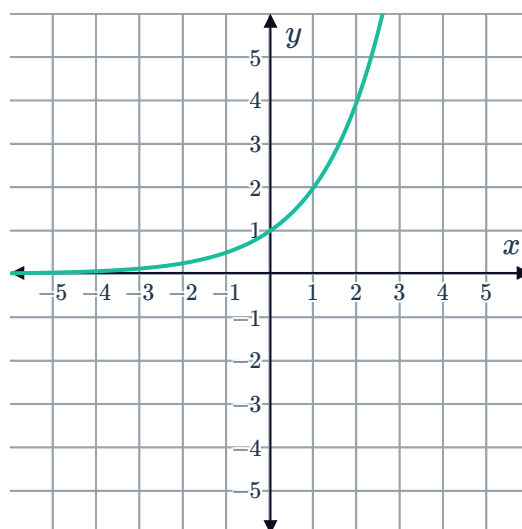
2 Consider the graph of  $y = \frac{1}{x}$ :

- a What translation is required to shift the graph of  $y = \frac{1}{x}$  to get the graph of  $y = \frac{1}{x} + 4$ .
- b Hence, sketch  $y = \frac{1}{x} + 4$ .

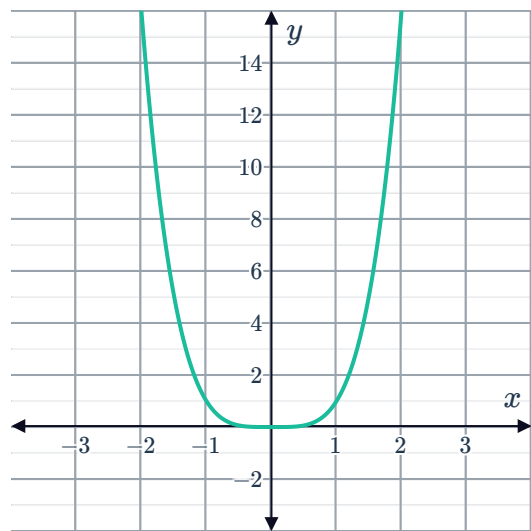


3 Consider the graph of  $y = 2^x$ :

- a What translation is required to shift the graph of  $y = 2^x$  to get the graph of  $y = 2^x - 5$ .
- b Hence, sketch  $y = 2^x - 5$ .



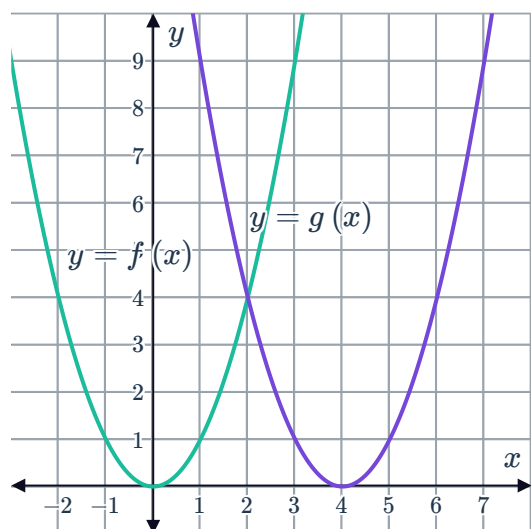
- 4 A graph of  $y = x^4$  is shown. Sketch the curve after it has undergone a transformation resulting in the function  $y = x^4 - 2$ .



- 5
- How should you translate the graph of  $y = f(x)$  to get the graph of  $y = f(x) + 4$ ?
  - How should you translate the graph of  $y = g(x)$  to get the graph of  $y = g(x + 6)$ ?

- 6 The functions  $f(x)$  and  $g(x) = f(x + k)$  have been graphed:

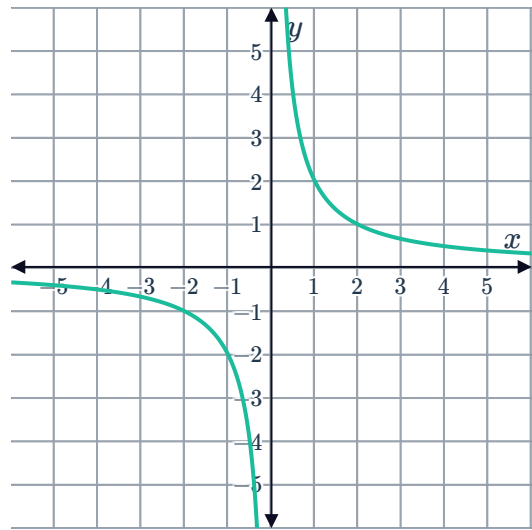
- Determine the value of  $k$ .
- Describe the transformation that occurred.



- 7 Describe the shift required to transform the graph of  $y = a^x$  to get the graph of  $y = a^{(x+5)}$ .
- 8 For each of the following equations, determine what the new equation is when their graphs are moved as described:
- The graph of  $y = x^3$  is moved to the right by 10 units.
  - The graph of  $y = 2^x$  is moved down by 9 units.

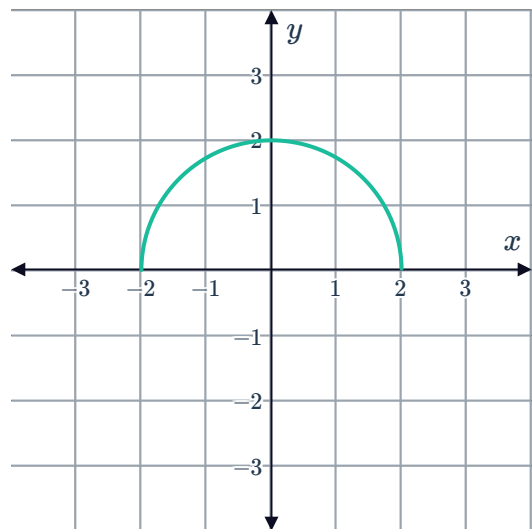
9 Consider the graph of the hyperbola  $y = \frac{1}{x}$  

- a What would be the new equation if the graph was shifted upwards by 2 units?
- b What would be the new equation if the graph was shifted to the right by 9 units?



10 Consider the graph of  $y = \sqrt{4 - x^2}$ :

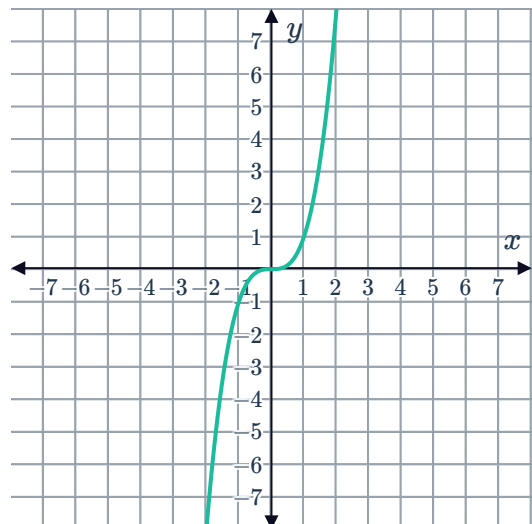
- a What would be the new equation if the graph was translated downwards by 7 units?
- b What would be the new equation if the graph was translated to the left by 3 units?



11 If the graph of  $y = x^4$  is moved to the right by 8 units and up by 6 units, what is its new equation?

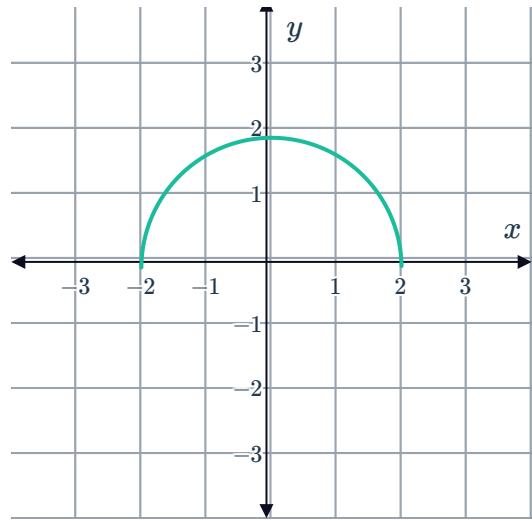
 12 Consider the graph of  $y = x^3$ :

- a Describe the required translations to shift the graph of  $y = x^3$  to get the graph of  $y = (x + 2)^3 + 4$ .
- b Hence, sketch  $y = (x + 2)^3 + 4$ .



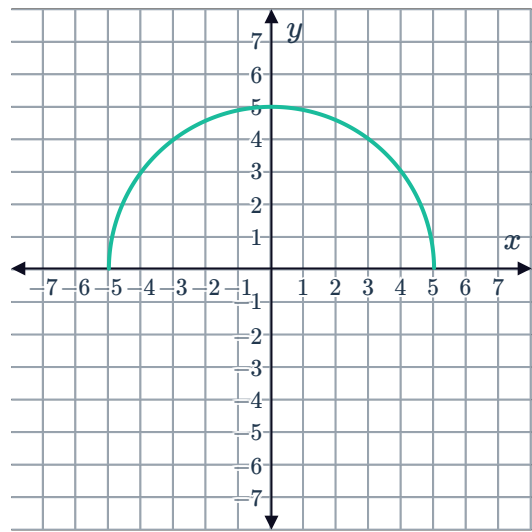


- a Describe the required translations to shift the graph of  $y = \sqrt{4 - x^2}$  to get the graph of  $y = \sqrt{4 - x^2} + 2$ .
- b Hence, sketch  $y = \sqrt{4 - x^2} + 2$ .

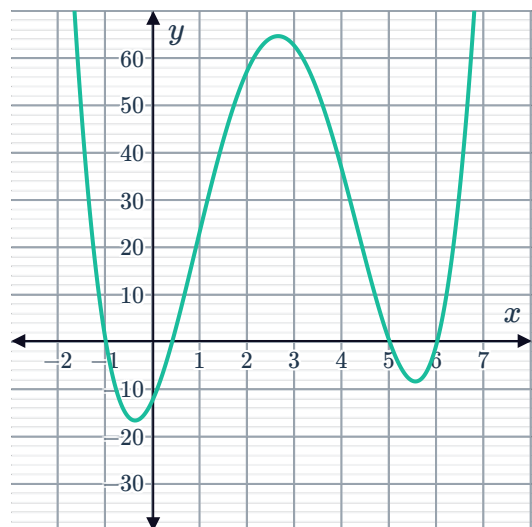


14 Consider the graph of  $y = \sqrt{25 - x^2}$ :

- a Describe the required translations to shift the graph of  $y = \sqrt{25 - x^2}$  to get the graph of  $y = \sqrt{25 - (x + 4)^2} - 2$ .
- b Hence, sketch  $y = \sqrt{25 - (x + 4)^2} - 2$ .



15 The graph of  $y = P(x)$  is shown. Sketch the graph of  $y = P(x) - 20$ .





## Reflections

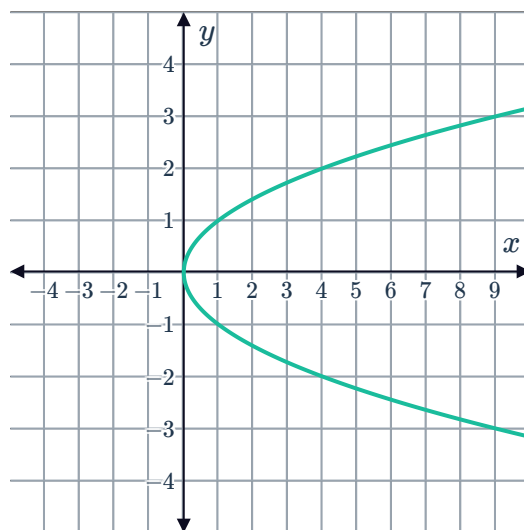
**16** Consider the point  $P(8, -3)$ . Sketch the point that is symmetric to  $P(8, -3)$  with respect to:

- a** The  $x$ -axis                      **b** The  $y$ -axis                      **c** The origin

**17** State whether the following functions have symmetry. If so, state the line or point of symmetry.

- a**  $y = x^2 + 1$                                               **b**  $y = x + 5$

**18** Is the graph shown symmetric with respect to the  $x$ -axis, the  $y$ -axis, or the origin?



**19** Does the function  $y = 4x^3$  have reflective symmetry?

**20** For each of the following functions:

- i** Find  $f(-x)$ .  
**ii** State whether the functions is symmetrical along the  $x$ -axis, the  $y$ -axis or neither.

- a**  $f(x) = 2x^3 - 2x$                                               **b**  $f(x) = 0.9x^6 - 7x^2 + 4$   
**c**  $f(x) = x^3 + x - 3$                                               **d**  $f(x) = \frac{1}{6x^3}$

**21** Consider the point  $P = (-6, 1)$ . Find the point obtained by:

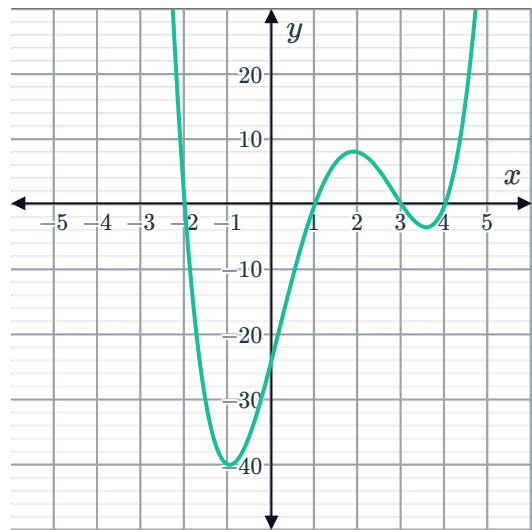
- a** Reflecting  $P$  about the  $x$ -axis                                              **b** Reflecting  $P$  about the  $y$ -axis  
**c** Rotating  $P$  by  $180^\circ$  about the origin

22 Describe the symmetry of the graph of  $x^2$   = 6.

23 Suppose  $f$  is a function such that  $f(3) = 2$ . State a point that lies on the graph of  $f$  if:

- a The graph of  $y = f(x)$  is symmetric with respect to the origin.
- b The graph of  $y = f(x)$  is symmetric with respect to the  $y$ -axis.
- c The graph of  $y = f(x)$  is symmetric with respect to the line  $x = 6$ .
- d  $f$  is an even function.
- e  $f$  is an odd function.

24 The graph of  $y = P(x)$  is shown. Sketch the graph of  $y = P(-x)$ .



## Dilations

25 Consider the function  $f(x) = \sqrt{x}$ . Write down the new function  $g(x)$  which results from scaling  $f(x)$  vertically by a factor of  $\frac{1}{3}$ , and scaling horizontally by a factor of  $\frac{1}{2}$ .

26 A function  $f(x)$  is transformed into a new function  $g(x) = f\left(\frac{x}{k}\right)$ . If  $0 < k < 1$ , determine what will be the effect on the graph of  $f(x)$ .

 **27** Consider the function  $f(x) = \sqrt{x}$ .

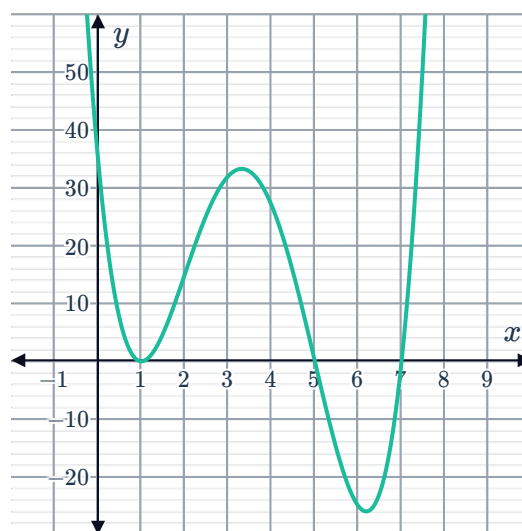


- a** Complete the table for  $f(x)$ :  
**b** Sketch a graph of the function  $f(x)$ .

|        |   |   |   |   |    |
|--------|---|---|---|---|----|
| $x$    | 0 | 1 | 4 | 9 | 16 |
| $f(x)$ |   |   |   |   |    |
| $g(x)$ |   |   |   |   |    |

- c** Complete the table of values for the transformed function  $g(x) = f(4x)$ , giving all values in exact form.  
**d** Sketch a graph of  $g(x)$ .  
**e** Describe how  $g(x)$  relates to the graph of  $f(x)$ .  
**f** Explain why the transformed function  $h(x) = f(-4x)$  can be undefined for the values of  $x$  in the table.

**28** The graph of  $y = P(x)$  is shown. Sketch the graph of  $y = 2P(x)$ .



## Combinations of transformations

 **29** Suppose that the  $x$ -intercepts of the graph of  $y = f(x)$  are  $-5$  and  $6$ . Find the  $x$ -intercepts of the graph of:

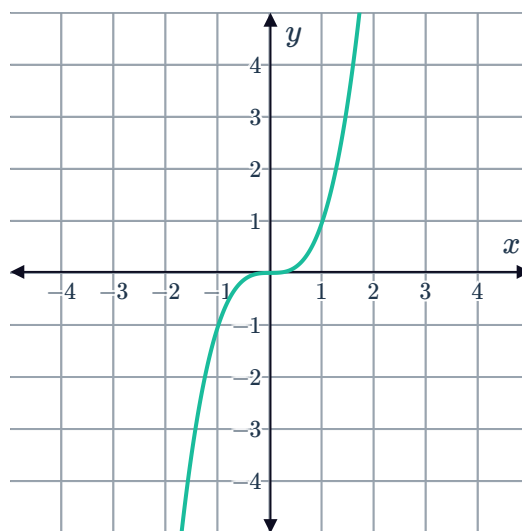
- a**  $y = f(x + 4)$       **b**  $y = f(x - 4)$       **c**  $y = 3f(x)$       **d**  $y = f(-x)$

**30** Consider the function  $f(x) = x^2 - 5$ . Using function notation, describe the transformation of  $f$  that will result in the function:

- a**  $g(x) = x^2 - 6$       **b**  $h(x) = 16x^2 - 5$       **c**  $k(x) = 3x^2 - 15$

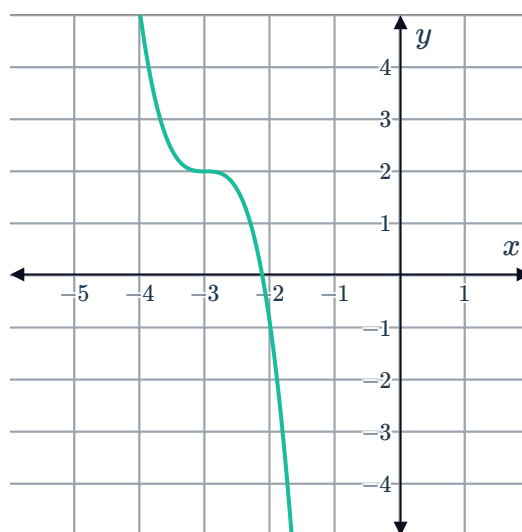


- 31** Consider the graph of  $y = x^3$ :  
Sketch the curve after it has undergone transformations resulting in the function  $y = -4(x + 4)^3$ .



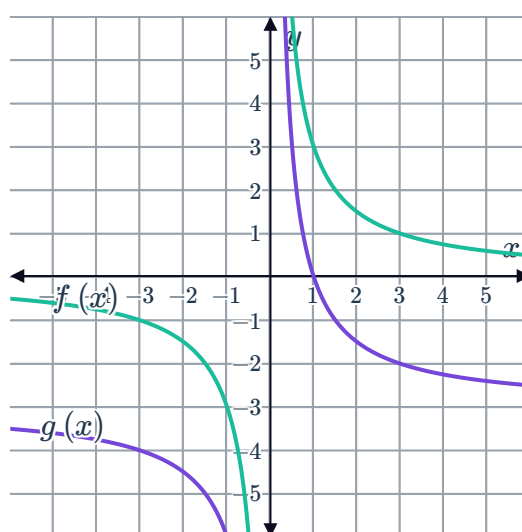
- 32** Consider the given graph:

- Describe the transformations of the graph of  $y = x^3$  to the given graph.
- Write down the equation of the given graph.



- 33** Consider the graphs of  $f(x) = \frac{3}{x}$  and  $g(x)$  shown:

- Write  $g(x)$  in terms of  $f(x)$  using the transformation shown in the graph.
- State the equation of  $g(x)$ .

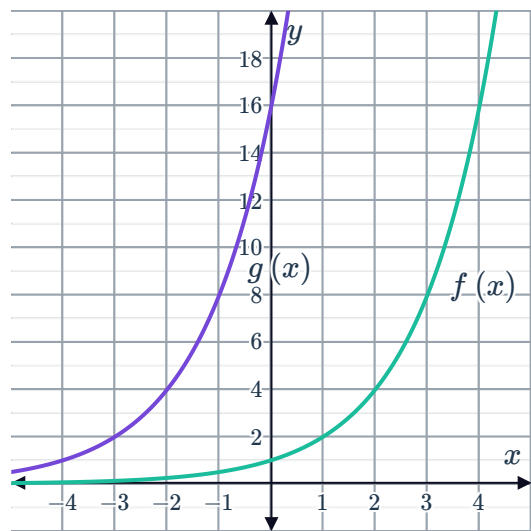




34 Consider the graphs of  $f(x) = 2^x$  and  $g(x)$  shown.



- Write  $g(x)$  in terms of  $f(x)$  using the transformation shown in the graph.
- State the equation of  $g(x)$ .



35 For the following base functions, describe the transformations that have occurred:

- From  $y = f(x)$  to  $y = f\left(\frac{x}{4} + 3\right) - 5$
- From  $y = g(x)$  to  $y = \frac{g(x - 3) + 3}{2}$
- From  $y = f(x)$  to  $y = 2f\left(\frac{1}{2}x\right) - 3$
- From  $y = h(x)$  to  $y = 2(h(x) + 3)$

36 For the following functions, describe the transformations that have occurred:

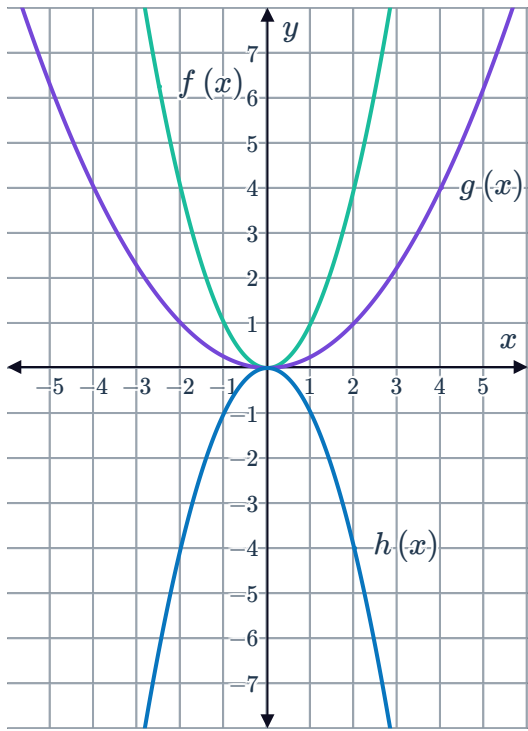
- From  $f(x) = x^4$  to  $g(x) = -7(x + 4)^4$
- From  $f(x) = x^4$  to  $g(x) = -5x^4 + 8$
- From  $f(x) = x^2$  to  $g(x) = -10(x + 8)^2 - 9$

37 The table below shows values that satisfy the function  $f(x) = |x|$ .

| $x$             | -3 | -2 | -1 | 0 | 1 | 2 | 3 |
|-----------------|----|----|----|---|---|---|---|
| $y = f(x)$      | 3  | 2  | 1  | 0 | 1 | 2 | 3 |
| $g(x) = 5f(x)$  |    |    |    |   |   |   |   |
| $h(x) = -2f(x)$ |    |    |    |   |   |   |   |

- Complete the table of values for each transformation of the function  $f(x)$ .
- Sketch the graph of  $g(x)$ .
- Describe how to transform the graph of  $f(x)$  into the graph of  $g(x)$ .
- Sketch the graph of  $h(x)$ .
- Describe how to transform the graph of  $f(x)$  into the graph of  $h(x)$ .

38 Three functions have been graphed on the number plane.  $g(x)$  and  $h(x)$  are both transformations of  $f(x)$ .



- a State the equation of  $f(x)$ .
- b Write  $g(x)$  in terms of  $f(x)$  using the transformation shown in the graph.
- c State the equation of  $g(x)$ .
- d Write  $h(x)$  in terms of  $f(x)$  using the transformation shown in the graph.
- e State the equation of  $h(x)$ .

39 Some points on the graph of the function  $y = f(x)$  are given in the table below:

| Original point | $g(x)$            | Corresponding point |
|----------------|-------------------|---------------------|
| $(9, -12)$     | $g(x) = f(x) - 8$ |                     |
| $(9, -12)$     | $g(x) = 6f(x)$    |                     |
| $(6, -7)$      | $g(x) = f(x - 5)$ |                     |

Complete the table by finding the corresponding points on the graph of  $y = g(x)$ .

40 Suppose that  $(-4, 3)$  is a point on the graph of  $y = g(x)$ . Find the corresponding point on the graph of:

- a  $y = g(x + 7) - 6$
- b  $y = -6g(x - 4) + 6$
- c  $y = g(6x + 1)$

41 The table below shows coordinates of points on the function  $y = f(x)$ . By performing the given transformation on them, find the corresponding transformed points.

| Point      | Transformation                      | Transformed point |
|------------|-------------------------------------|-------------------|
| $(-3, -1)$ | $y = f(x - 5) - 2$                  |                   |
| $(0, 3)$   | $y = \frac{1}{5}f(x - 3)$           |                   |
| $(1, -2)$  | $y = f(3x) - 2$                     |                   |
| $(-3, 5)$  | $y = f\left(\frac{x}{2}\right) - 5$ |                   |